Md. Jaied Hasan

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Academic Credentials

Bangladesh University of Engineering and Technology (BUET), Dhaka Jan 2022 – Jun 2026
Bachelor of Science in Electrical and Electronic Engineering **CGPA: 3.80/4.00** (Top 27%)
Major: Communication and Signal Processing *Last four semesters: 3.85/4.00*

Experience

FlyRank AI Aug 1, 2026 – Present
Machine Learning Intern *Remote*

- Developed an ML pipeline to detect and rank declining web content across 23,000+ webpages using historical traffic and trend-based features.
- Performed feature engineering, Precision@K evaluation, error analysis, and model diagnostics to translate predictions into actionable content-review recommendations.
- Strengthened model reliability through time/group-aware validation, data-leakage audits, and failure analysis, aligning reported conclusions with experimentally supported evidence.

Banglalink Digital Ltd. 18 Nov 2025 – 25 Nov 2025
Industrial Attachment

Bangladesh Mathematical Olympiad Nov 2022 – Dec 2025
Academic Team Member

Research Experience

Bangladesh University of Engineering and Technology (BUET), EEE Sep 15, 2026 – Present
Research Assistant | Supervisor: Dr. Md. Forkan Uddin, Professor, EEE, BUET

- Conducting research on Wi-Fi 8 (IEEE 802.11bn) multi-AP coordination, with emphasis on coordinated beamforming (Co-BF), multi-AP optimization, and performance under inter-cluster interference (ICI).
- Developed a Python-based multi-AP simulator to compare three operating modes: conventional CSMA/CA with beamforming, Co-BF with ICI, and Co-BF without ICI.
- Investigating AP recruitment/pairing and multi-AP coordination strategies, with particular focus on how ICI changes the throughput and interference behavior of coordinated deployments.

Undergraduate Thesis: Federated Sparse PCA with Differential Privacy Using Correlated Noise
Supervisor: Dr. Hafiz Imtiaz, Professor, EEE, BUET

- Developed a federated sparse-PCA framework for horizontally partitioned datasets.
- Investigated correlated-noise mechanisms for differentially private dimensionality reduction and feature selection.
- Evaluated privacy–utility, convergence, sparsity, reconstruction, and communication-efficiency trade-offs across distributed settings.
- Implemented reproducible experimental pipelines in Python using NumPy, Pandas, and scientific-computing tools.
- Applied linear algebra, optimization, statistical analysis, and algorithmic evaluation to privacy-preserving distributed learning.

Publications

- **Journal manuscript under review (first author):** "Federated Sparse PCA with Differential Privacy Using a Correlated-Noise Mechanism." *The Journal of the Franklin Institute* (Q1).
- **Conference paper accepted (2026; first author):** "An Integrated Dual-Module Cyber Physical System for Motorcycle Safety: Utilizing Pre-Ride Safety, Crash Detection and Swift Response with Mobile Application Companion." *IEEE ICACCESS 2026*(Acceptance rate 23.3%).
- **Conference submission (2026):** "ML-Based Mobile Traffic Forecasting for Network-Capacity Planning." Submitted to *IEEE ICECE 2026*.

Technical Skills

Programming	Python, C, C++, MATLAB, Verilog
ML / Deep Learning	PyTorch, TensorFlow, Keras, Scikit-learn, XLM-RoBERTa, MobileNetV2, KNN
Trustworthy / Efficient AI	Federated learning, differential privacy, sparse PCA, knowledge distillation, class-weighted learning
Signal / Data Processing	Digital signal processing, MFCC, time-series analysis/forecasting, PCA/SPCA, NumPy, Pandas
Wireless / Networking	Wi-Fi 8 (IEEE 802.11bn), multi-AP coordination, coordinated beamforming (Co-BF), inter-cluster interference analysis, NAV modeling, angle-based channel modeling, beam-pattern/SINR analysis, wireless-network optimization
Embedded / IoT / Edge	ESP32, Arduino, Raspberry Pi, ESP32-CAM, IMU/MPU6050, GPS, GSM/SIM800L, embedded sensor integration
Protocols	I2C, SPI, UART
Software / Engineering Tools	Linux, Git/GitHub, ROS, MATLAB/Simulink, Proteus, PSpice, AutoCAD, KiCad, MIT App Inventor

Selected Projects

Multilingual Toxicity Classification with Knowledge Distillation
HerWILL AI for Digital Safety Datathon 2026

- Developed an XLM-RoBERTa teacher–student pipeline for Bangla, English, and Banglish/code-mixed harmful-content classification.
- Fine-tuned the teacher model on a 47K+ sample, three-class dataset using stratified 5-fold training; variants document class-imbalance handling, label smoothing/class weighting, mixed precision, temperature scaling, and macro-F1 calibration.
- Designed a compact approximately 5M-parameter student with a custom byte-level BPE tokenizer and mean/max pooling, using temperature-scaled teacher probabilities for knowledge distillation.
- Achieved approximately 55× parameter compression relative to the teacher; the research CV records a 0.344 F1NOP score for the student pipeline.

HIVE-R / GPS-Free Rescue Drone for Disaster Response

- Developed a GPS-independent semi-autonomous search-and-rescue drone for post-disaster environments, with work spanning IMU-assisted inertial navigation and a dual-camera visual-odometry pipeline for GPS-denied pose estimation.
- Deployed a human-detection model on Raspberry Pi for onboard victim identification and integrated reporting to a ground-control station over a wireless link.
- Worked on embedded sensing, edge computation, autonomous navigation, sensor integration, and hardware–software integration for field-oriented robotic operation.

LifeSaver Wearable Smart Helmet

- Developed an ESP32-based IoT smart helmet integrating MPU6050/IMU-based fall detection, SIM800L GSM emergency communication, GPS localization, alcohol sensing, and overspeed monitoring.
- Implemented a two-module helmet/bike architecture with ESP32–ESP8266 communication, helmet-wear and alcohol checks, GSM/GPS emergency messaging, and a Bluetooth-connected Android companion application.

Autonomous Fire-Extinguishing Robot

- Developed an ESP32- and Arduino-based mobile robot integrating flame sensors, L298N motor drivers, autonomous navigation, and fire-suppression actuation.
- Used ESP-NOW for inter-controller communication; implemented flame-directed motion, fan/sprayer actuation, and a temperature-triggered retreat rule in a two-room test environment.

ML-Based Mobile Traffic Forecasting and Dynamic Bandwidth Planning

- Developed a forecasting pipeline for regional mobile-internet traffic using BTCL data, with emphasis on the Savar region.

- Benchmarked XGBoost, GRU, and LSTM architectures using forecasting metrics including MAE and RMSE to study network-demand prediction and data-driven bandwidth/capacity planning.
- A paper based on this work was submitted to a local conference in 2026.

Lung Disease Classification from Respiratory Audio

- Built a pulmonary-disorder classification system in MATLAB using respiratory sound signals.
- Applied Mel-Frequency Cepstral Coefficients (MFCC) for feature extraction and K-Nearest Neighbors (KNN) for multiclass disease classification; one project variant reports 98.66% macro sensitivity.
- Evaluated the respiratory-audio pipeline on the ICBHI respiratory dataset across multiple pulmonary-disorder categories.

ESP-Eye: Embedded Vision System

- Developed an edge-vision platform integrating an ESP32-CAM video stream with a pretrained MobileNetV2 model for real-time object recognition.
- Implemented live prediction labels and confidence-score overlays, exploring lightweight vision deployment under constrained embedded-computing conditions.

Dual-Channel Pressure Monitoring System

- Implemented a dual-channel pressure-monitoring device using an ESP32 microcontroller.
- Streamed real-time pressure data to a web dashboard while supporting local LCD/button control and remote monitoring/configuration.

Wi-Fi 8 Multi-AP Coordinated Beamforming Simulator

- Developed a Python simulation codebase to study multi-AP coordinated beamforming and inter-cluster interference, complementing ongoing BUET research on IEEE 802.11bn.
- Compared CSMA/CA-based beamforming and coordinated transmission scenarios, with channel, precoding, interference, and per-AP throughput analysis.

ESP32 Smart Voltage and Overcurrent Protection System

- Developed an ESP32-based embedded voltage and overcurrent protection project organized around finite-state-machine (FSM) logic for monitoring and protective decision-making.

Attendance System (Python)

- Developed a Python-based attendance-system project and maintained its implementation in a public source-code repository.

Machine Learning Content-Performance Analysis – Project Repository

- Maintained a Jupyter Notebook codebase related to the FlyRank AI internship, including work on content-performance trends, feature engineering, model assessment, and error analysis.

PCB Design Collection

- Organized and published personally designed printed circuit board (PCB) layouts and related hardware design artifacts in a dedicated repository.

Digital-Logic Tic-Tac-Toe

- Implemented a hardware Tic-Tac-Toe game using basic logic-gate and flip-flop integrated circuits and LED-based game-state indicators.

Additional Engineering Projects

- Transient-analysis application.
- FDM project.
- Frequency-converter project.

Research and Technical Interests

- Trustworthy and Privacy-Preserving ML
- Federated and Distributed Learning
- Efficient and Scalable AI
- Robust Machine Learning
- Multimodal and Foundation Models
- AI for Healthcare
- Deep Learning
- Signal Acquisition and Processing
- Wi-Fi 8 (IEEE 802.11bn) and Multi-AP Coordination
- Coordinated Beamforming (Co-BF) and Interference-Aware WLAN Optimization
- Wireless Communications and Network Optimization
- Directional Channel and Beamforming Analysis
- Wireless Networks, IoT, and Edge Intelligence
- Data-Driven Communication Systems
- Cyber-Physical Systems
- Robotics and Automation
- Mechatronics and Embedded Systems
- Human–Robot Interaction
- Embedded Systems and Edge Devices

Relevant Coursework

- Artificial Intelligence and Machine Learning
- Machine Learning Specialization (Coursera)
- Digital Signal Processing
- Introduction to Medical Imaging
- Probability and Statistics
- Wireless Communications
- Communication Systems I and II
- Microprocessor and Embedded Systems
- Robotics and Automation
- Control Systems
- Electronic Circuits I and II
- Digital Electronics
- DC/AC Circuits
- Power Electronics
- Structured Programming in C/C++

Honors and Awards

- **Dean's List Scholarship** – awarded for excellent academic performance in Level 1 and Level 3.
- **University Merit Scholarship** – awarded for excellent academic performance.
- **Shamsul Huda Chowdhury Scholarship** – BUET trust-fund scholarship for students with outstanding academic records.
- **Professor Dr. Ahmad Ullah and Mrs. Chemanara Ahmad Ullah Trust Fund Scholarship** – awarded for outstanding academic performance at BUET.
- **1st Runner-Up, Brain of Varendra** – talent-hunt competition organized by Varendra University, Rajshahi.